

Pin Usage — ESP32-S3 Feather (Adafruit 5477)

Arduino variant: adafruit_feather_esp32s3

I2C Bus

SDA = GPIO 3 (A6) | SCL = GPIO 4 (A7) | Pull-ups: 5 kΩ on-board + STEMMA QT

Address	Device	Product
0x3C	SH1107 OLED	Adafruit 4650 FeatherWing
0x51	DimmerLink – heat	RBDimmer DimmerLink
0x52	DimmerLink – fan	RBDimmer DimmerLink
0x76/77	BME688 env sensor (future)	Adafruit 5046 via STEMMA QT

SPI Bus

SCK = GPIO 36 | MOSI = GPIO 35 | MISO = GPIO 37

GPIO	Feather label	Firmware define	Device	Role
36	SCK	–	shared	SPI clock
35	MOSI	–	shared	SPI data out
37	MISO	–	shared	SPI data in
10	D10/A11	CS_RTD_BT	MAX31865 (Adafruit 3648)	Bean temp PT1000
9	D9/A10	CS_RTD_ET	MAX31865 (Adafruit 3648)	Exhaust/ET PT1000
8	A5	CS_TC_IN	MAX31855 (Adafruit 269)	Inlet K-type thermocouple

Reserved — Do Not Use

GPIO	Label	Reason
3	SDA / A6	I2C in use
4	SCL / A7	I2C in use
7	–	I2C power control (board-managed)
13	D13	Built-in red LED
21	–	NeoPixel power enable
33	–	NeoPixel data
35	MOSI	SPI in use
36	SCK	SPI in use
37	MISO	SPI in use

Free GPIOs

GPIO	Feather label	GPIO	Feather label
5	A8	6	A9
11	A12	12	A13
14	A4	15	A3
16	A2	17	A1
18	A0	38	RX / Serial1
39	TX / Serial1	42	SS (default SPI SS)